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NWR-138

NORTHROP AIRCRAFT, INC.
Hawthorne, California

MONTHLY PROGRESS REPORT
April 22, 1948

NEPA 583-NOR-27

Reference: NEPA Research Program (Our S.O. 3078)

Subject: Generalized Study of the Application of Unconventional
Power Plants to All-Wing Type Aircraft

Submitted by: H. E. Metcalf and W. J. Cerny

342-75-0137

Box 12,
File 17INTRODUCTION:

The reader is referred to the previous Monthly Progress Reports for the general objectives of the program and the progress through March 22, 1948.

POWER PLANT STUDIES (NORTHROP-HENDY):

Except for some liaison work on installation and ducting problems, the design work of Northrop-Hendy is considered complete.

Below are summarized the performance data for the turbo-jets as determined by Northrop-Hendy. A portion of this performance data is a repetition of previously reported data. Note that compressor lengths have been changed from those values shown on the Monthly Progress Report for March 22, 1948. Earlier figures were based on estimates. More detailed analysis necessitated the change to the higher figures shown here.

PERFORMANCEThrust @ 35,000 Ft. Altitude

<u>Unit</u>				Air Flow @ S.L. Static
	400 MPH	500 MPH	600 MPH	
10,000# Thrust @ S.L.	3,550	3,630	3,720	174 #/sec.
20,000# " " "	7,100	7,260	7,440	348 "
30,000# " " "	10,650	10,890	11,116	525 "

Ducting Dimensions and Weight

Unit		Intake	Comp.		Duct to Turbine		Tailpipe Dia., In.
		Duct	Disch.	Duct	Area	Wt.	
		Area	Area	Wt.			
		Sq.Ft.	Sq.Ft.	#/Ft.			
10,000# Thrust @ S.L.		5.03	0.75	24	1.25	47	31
20,000# " " "		10.06	1.50	34	2.49	66	44
30,000# " " "		15.1	2.25	42	3.74	81	54

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April 16, 1947

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Dr. Jerome C. Hunsaker, Chairman,
National Advisory Committee for Aeronautics,
1500 New Hampshire Ave., N. W.,
Washington 25, D. C.

Dear Dr. Hunsaker:

At the meeting of the National Advisory Committee for Aeronautics on January 24 you asked us to give consideration to a suggestion from Admiral Stevens, in his letter of January 10, that the NACA establish a committee on the application of nuclear power for aircraft propulsion. I have done this as best I could in view of the uncertainties which have confronted the Atomic Energy Commission and feel that it might be well to make a report to the NACA at this time. The situation may be summarized as follows:

1. The Atomic Energy Commission, although interested broadly in development of power from nuclear sources, seems not to have projected any work on the specific field of application to aircraft propulsion.

2. Some preliminary orientation studies on the question have been made by Dr. Harvey Hall, technical aide to Admiral Stevens, and by Dr. Arthur Ruark of the Applied Physics Laboratory of Johns Hopkins University, working for Navy Bureau of Aeronautics.

3. Some general studies on the thermodynamic and heat transfer problems of reactor application have been started at the NACA Aircraft Engine Laboratory at Cleveland.

4. The Army Air Forces have established the NEPA (Nuclear Energy for Propulsion of Aircraft) project with headquarters at Oak Ridge, Tenn., for a study of this subject. This is established by a prime contract to the Fairchild Engine and Airplane Corporation, but has established consulting arrangements with all the major aircraft engine manufacturing companies in this country. The NACA group at Cleveland has already established working relations with the NEPA project. At present the NEPA project is the only one in this field to which the Atomic Energy Commission has authorized the transmittal of classified information.

5. It seems to me that the NACA should actively assume responsibility for a continuing study of this subject. There is every indication at present that the Atomic Energy Commission intends to decentralize by assigning special areas of responsibility to contractors, so it is assumed that the AEC will welcome NACA cooperation in this field of mutual interest.

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6. Because of the special importance of the project and the very special security restrictions associated with it, it is recommended that NACA have a special committee on nuclear energy for aircraft propulsion. This recommendation differs from the original suggestion of Admiral Stevens that the subject be handled as a Subcommittee under the Power Plant Committee.

In a letter from Major General Curtis LeMay of the Army Air Forces to the writer, on this subject, dated March 4 he says that "this office feels that from a policy standpoint such a function is essentially now in hand via participation by the NACA group on the NEPA project." He further writes:

"In view of the above, it is the opinion of this office that the N.A.C.A. might best function in the field of nuclear research particularly applicable to aircraft and guided missiles propulsion by continuing in its present capacity as a participating agency of the NEPA Division. The establishment of a new and separate N.A.C.A. committee or group to pursue such work would, in essence, duplicate to a large degree authority and responsibility already vested in the A.A.F. and, insofar as is presently understood, would be contrary to the desires of the Atomic Energy Commission. In any event, if, after having had the opportunity of reviewing information contained herein, you are of the opinion that the N.A.C.A. should establish a special nuclear energy committee, it is suggested that your views and recommendations be referred to the N.A.C.A. committee of which you are a member, for consideration and decision."

It seems to me that the subject is of such profound interest to NACA that NACA should set up a committee to consider it from the NACA standpoint. Such a committee would of course avail itself fully of the cordial invitation of Gen. LeMay for us to cooperate with the NEPA project. In so doing it should be possible to avoid unnecessary duplication of effort.

7. Some thought was given to possible make-up of such a committee and the following seemed reasonable:

Chairman, to be appointed from the regular membership of the NACA.

Members from the following suggested list:

A representative from the technical staffs of the War Department, the Navy Department and the Atomic Energy Commission; Mr. A. H. Rothrock, director of research of the NACA Cleveland Laboratory and Dr. Simon of NACA Cleveland who is now assigned to work at the Clinton Laboratory in Oak Ridge; Prof. McAdams of the M.I.T. for his experience in heat transfer; Dr. John Chipman, M.I.T., Dr. Cyril S. Smith of University of Chicago and Dr. L. O. Brockway, University of Michigan.

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for metallurgy; Dr. McCullough or Dr. Daniels of the Clinton Laboratory for prior experience in high temperature reactor design; Dr. Zinn of the Argonne Laboratory and Dr. P. M. Morse of the Brookhaven Laboratory, for cooperation with these two major basic science laboratories of the Atomic Energy Commission.

8. If such a committee is appointed, its first duty should be to review the ideas already developed by the Cleveland NACA group. At the same time the Chairman of NACA should officially notify the Atomic Energy Commission of the establishment of the Committee and take steps to have its members cleared for access to classified material. In this way direct liaison would be established with the Divisions of Research and Engineering of the Atomic Energy Commission. Concurrently, the committee should establish formal working arrangements with the NEPA project and with the Clinton Laboratories at Oak Ridge.

Sincerely yours,

E. U. Condon

E. U. Condon, Director.

NEPA DIVISION

cc: Dr. A. M. Weinberg
Prof F. Wheeler Loomis
Mr. T. A. Sims (3)
Mr. G. Knepf
Subj. File
Date File

March 9, 1950

Mr. George S. Schairer
Boeing Airplane Company
Seattle 14, Washington

Dear Mr. Schairer:

I am writing to you in connection with an important problem which faces us here at the NEPA Project. As you may by now have heard, the four government agencies which constitute the Aircraft Nuclear Propulsion Program are sponsoring a Summer Session in Oak Ridge during the Summer of 1950, the purpose of which is to assist the various working groups in defining and solving the problems of Nuclear Powered Flight. I am enclosing for your information a copy of a rough charter for this summer working session which I drew up some weeks ago.

We have been able to get Professor F. Wheeler Loomis, Head of the Physics Department of the University of Illinois, to act as the chairman of the session for this summer. Professor Loomis was a senior member of the Lexington Project and is well known in physics. We are also finding it possible to secure the assistance of a number of other highly qualified individuals in various branches of science. In order, however, to have a well rounded group it will, of course, be necessary that it contain not only a strong scientific body, but also that it contain at least a few persons who are acquainted with the practical aspects of designing, building and operating large military aircraft. Unless the group contains such a person, it well might turn out to be more of a hindrance than a help to the Aircraft Nuclear Propulsion Program, through lack of balance and through failure to attach adequate importance to the many practical problems which lie between the ANP Project at present and the ultimate realization of a nuclear powered aircraft. In our thinking and discussions, your name repeatedly comes up because of your experience and qualifications in this field. I realize that you have many urgent and important duties in connection with your work at Boeing and we are, of course, well aware of the importance of the work which your company is doing. However, the kind of mature advice and assistance which you can give is so badly needed and so crucially important for the ANP Project that I feel justified in asking you to spend about two months here in Oak Ridge this summer on this Project.

Mr. George S. Schairer

March 9, 1950

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You may wish to know something of the standing which the summer working session has with the government agencies which sponsor the ANP Program. I can say that the summer working session is regarded by them as a matter of very great importance for the ANP Program. When we were trying to secure the services of Professor Loomis, Dr. L. R. Hafstad made a special trip to Urbana, Illinois, to interview Dr. Loomis, with A. M. Weinberg and me. Dr. K. T. Compton, formerly of MIT and former Chairman of the Research and Development Board, and Dr. James R. Killian, now President of MIT, both have indicated their wholehearted support of the idea and many people from MIT will take part in the session. Mr. Sims has talked with General Putt regarding it, and General Putt indicated that he also is strongly in favor of the working session and was, as a matter of fact, one of the several people who have suggested your participation in it. Some other arrangements regarding the working session will be of interest to you also. The members of the working session and their families will be offered new, and newly furnished, apartments here in Oak Ridge for the duration of their stay at a moderate rental. Costs of transporting members of the group and their families to and from Oak Ridge will also be paid. Salary arrangements will compare favorably with those which you normally have. Office space will be provided in the Y-12 Area here in Oak Ridge and the offices will be air-conditioned. In fact, I can assure you that everything will be done to make your stay both highly productive and also as pleasant as possible. Prof. Loomis' present plans call for a beginning of the session early in the month of June and a continuation until about the first of August. Following this, it is supposed that some members of the session may wish to continue their work here at Oak Ridge, but the main effort will by that time have been completed. You will notice that the summer working session is spoken of in the attached charter as a permanent and recurring institution. Your acceptance of duty with this session this summer does not, of course, necessarily bind you for similar duty in future years.

You may wish to know why Professor Loomis has not written to you directly. In our discussions he has expressed himself as being somewhat unfamiliar with the technical community in your field and thought that it would be more appropriate for us to approach you.

I sincerely hope that you will find it possible to accede to this request and I shall be awaiting your answer with the utmost interest.

Sincerely yours,

NEPA DIVISION
FAIRCHILD ENGINE & AIRPLANE CORP.

(S) M. C. Leverett
Technical Director

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